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A Mini Project Report (BIS586)

on

“Smart Department Portal”

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CERTIFICATE

This is to Certify that the Mini Project work on the topic “**Smart Department Portal**” has been successfully presented at **Don Bosco Institute of technology** by **Adithya K(1DB23IS007)**, **Jaishree K S(1DB23IS064)** in partial fulfillment of the requirements for the *V Semester degree of Bachelor of Engineering in Information Science and Engineering* of *Visvesvaraya Technological University, Belagavi* during academic year 2025-2026. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Mini project report deposited in the departmental library. The Mini Project report has been approved as it satisfies the academic requirements in respect of Mini Project work for the said degree.

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ABSTRACT

The rapid growth of educational institutions and digital learning environments has created a pressing need for unified academic management systems that streamline communication, automate repetitive tasks, and enhance the overall learning experience. Traditional paper-based processes such as attendance tracking, assignment handling, material distribution, and internal marks management are inefficient, prone to error, and difficult to maintain. To address these gaps, ISEHub : Integrated Student–Faculty–Admin Academic Portal has been developed as a modern, lightweight, and user-friendly web application designed to simplify academic operations within an educational institution. This project provides three role-based dashboards—Admin, Faculty, and Student—each offering customized tools based on user permissions. The Admin Dashboard enables centralized academic control including student management, internal marks entry, attendance updates, study material uploads, assignment uploads, and notification broadcasting. Admin has the highest privilege, ensuring authentication-secured access to sensitive academic data. The Faculty Dashboard supports instructors by allowing them to view students of their assigned sections, upload study materials, and update internal marks for subjects they handle. This ensures decentralized yet controlled management of academic information, reducing dependency solely on the administrative team. The Student Dashboard is designed with a clean, modern interface where students can view their attendance, internal marks, assignments, study materials, notifications, and timetable. Students can also submit assignments directly through the portal, making the system interactive and paperless. A key highlight of this project is the use of JSON files as a lightweight database, which eliminates the need for complex database servers while still providing structured data storage. All user-specific information—such as student records, faculty details, assignments, notes, notifications, and submissions—is managed through dedicated JSON files. The backend is implemented using Python Flask, a micro web framework that provides simplicity, scalability, and smooth routing mechanisms. Various backend functions, including authentication, file handling, session management, and data retrieval, are implemented efficiently to create seamless interactions between the frontend and backend components.

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Chapter 1

INTRODUCTION

1.1 Preamble

The rapid digital transformation in educational institutions has highlighted the necessity for centralized and automated academic management systems. Conventional approaches to managing attendance, internal marks, assignments, study materials, and notifications rely heavily on manual effort. These traditional workflows are time-consuming, lack transparency, and increase the likelihood of human error. As the academic ecosystem grows more complex, the need for an integrated and intelligent platform becomes essential to support efficient academic operations.

To overcome these challenges, the ISEHub Academic Portal was developed as an all-in-one solution that facilitates seamless academic coordination between students, faculty, and administrators. The backend of the application uses Flask, a lightweight yet powerful Python web framework, capable of handling dynamic routing and rendering web templates efficiently.

- Administrators can manage global operations including student and faculty databases, learning materials, attendance registry, internal marks, assignment distribution, result management, and general notifications.
- Faculty members can record and update attendance, upload study materials, assign and evaluate assignments, enter internal marks, and monitor student academic performance on a subject-wise basis.
- Students benefit from simplified access to their academic profile, where they can view attendance status, internal marks, notifications, timetable, study materials, and submit assignments digitally without physical interaction.

Overall, the ISEHub Academic Portal provides a modern, efficient, and user-friendly academic management system that aligns with the goals of digital education. It streamlines key administrative activities, helps faculty manage classroom data effortlessly, and empowers students with transparent access to academic information. The platform showcases how technology can enhance academic workflows, support paperless administration, and significantly improve the teaching-learning experience in higher education.

1.2 Problem Statement

Many educational institutions still depend on manual or partially digital methods to manage attendance, internal marks, assignments, and study materials. As a result, academic information remains scattered across WhatsApp groups, handwritten notes, spreadsheets, and emails, leading to miscommunication, delays, and data inconsistency. Students often need to personally contact faculty to check attendance or marks, while faculty struggle with maintaining multiple registers, updating spreadsheets, and repeatedly sharing materials. The absence of a centralized system causes confusion, errors, and poor coordination between students, faculty, and administrators. This study aims to address the following problems:

1. What is the current level of efficiency and accuracy in maintaining academic records through manual or semi-manual methods?
2. What challenges do students and faculty face in accessing attendance, internal marks, assignments, and study materials?
3. H₀: There is no significant improvement in coordination and data management when moving from a manual system to a centralized online academic portal.

1.3 Aim and Objectives of the Proposed Work

Aim:

Design and develop a centralized web-based academic management system that streamlines communication and coordination between students, faculty, and administrators by digitally managing attendance, internal marks, assignments, study materials, and notifications in a unified and user-friendly platform.

Objectives:

- To digitize and streamline the management of attendance, internal marks, and assignments.
- To provide students with real-time access to academic information such as notes, submissions, marks, and notifications.
- To enable faculty to efficiently upload study materials and manage student records.
- To minimize errors caused by manual record-keeping.
- To ensure secure, role-based login and access control for Admin, Faculty, and Students.
- To improve communication and reduce dependency on manual or external platforms.
- To minimize errors caused by manual record-keeping.

1.4 Proposed System

The proposed system is a unified digital platform that automates and streamlines daily academic activities within the department. Instead of handling assignments, attendance, notes, and announcements separately, the portal centralizes everything for Admins, Faculty, and Students. Admins can upload study materials, manage students, track internal marks, and post announcements. Faculty can maintain subject-related content and monitor student progress without relying on paper records. Students receive an intuitive dashboard to access assignments, notes, timetables, attendance, internal marks, and notifications in one place.

Built using Flask with JSON-based storage, the system ensures fast, accurate, and secure data handling through role-based authentication. The UI features soft pastel themes, interactive cards, and smooth navigation for users of all technical levels. By digitizing academic workflows, the platform reduces administrative effort, minimizes delays, and provides students with instant access to essential resources, improving overall academic efficiency.

1.4.1 Block Diagram of Proposed System

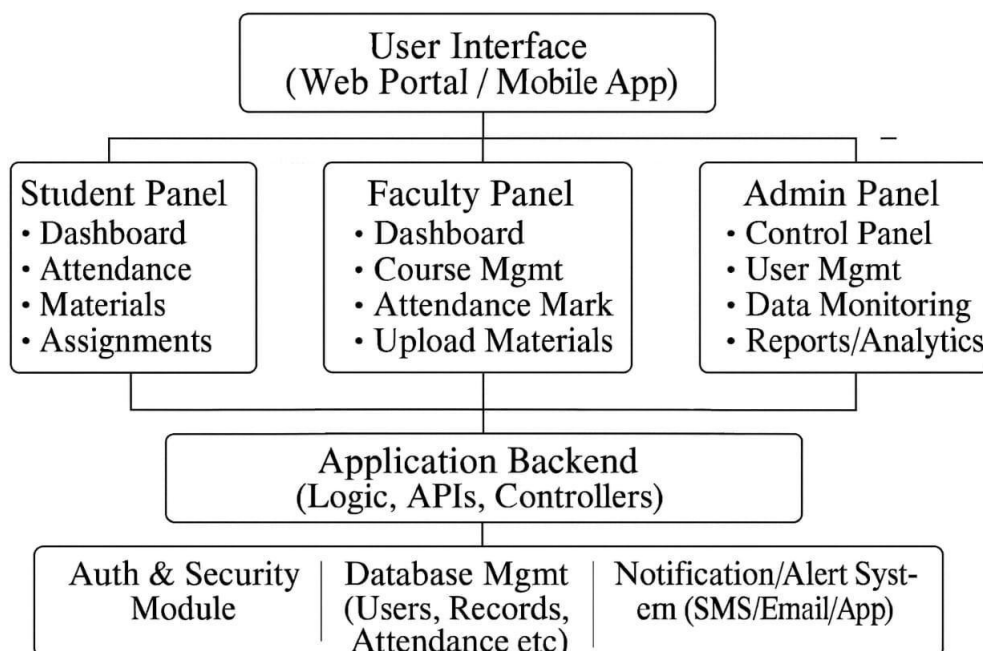


Figure 1.1: Proposed System Architecture Block diagram

The Smart Department Portal starts with a user-friendly interface accessible via web or mobile, allowing all users to log in, navigate dashboards, access materials, and receive notifications with ease. Behind this interface, the system is divided into three user panels—Student, Faculty, and Admin—each tailored to specific needs. The Student Panel provides access to attendance, study materials, assignments, and academic performance. The Faculty Panel enables teachers to upload materials, record attendance, manage courses, and monitor student progress. The Admin Panel offers full system control, including user management, data monitoring, and configuration.

1.4.2 Advantages of Proposed System

Centralized Academic Management

All academic activities—assignments, attendance, notes, marks, timetable, and notifications—are available in one platform, eliminating fragmentation.

Real-Time Access to Information

Students, faculty, and admin can instantly access updated academic data, improving transparency and reducing delays.

Reduced Manual Workload

Automated uploading, updating, and tracking minimize manual paperwork and administrative overhead.

Better Communication

Notifications and updates are delivered instantly to students, reducing miscommunication between faculty and learners.

Secure and Role-Based Access

Different login portals ensure that Admin, Faculty, and Students only access features meant for their roles, improving data safety.

Improved Student Engagement

Students can track their assignments, marks, attendance, and study materials easily, making academics more organized.

Modern & User-Friendly Interface

Pastel-themed dashboards with attractive animations make the system intuitive and enjoyable .

Chapter 2

Literature Survey

Table 2.1 Literature Survey

Paper	Year	Focus	Tech used / Datasets Use	Strengths	Limitations
[1]“Web Portals for Higher Education Institutions: A Review”	2018	Review of web portals in higher-education	Modular architecture, centralized portal design	Improved academic workflow, transparency, communication	Lacks personalization and adaptive systems
[2]Pavithra et al., “Department Information System – Web Application”	2019	Department-level digital academic information system	Web application, centralized DB	Enhances student access & reduces paperwork	Limited automation & no intelligent analytics
[3]Kogotkova et al., “Smart Department: Designing Digital Applications for Smart Organizations”	2021	Smart workflow for departments	Intelligent automation and process workflows	Secure data handling, scalability	Requires complex deployment & continuous updates
[4]Gonzalez-Longatt et al., “Secured Student Portal Using Cloud”	2021	Cloud-based student academic portal	Cloud infrastructure, encryption	High security & accessibility	Needs strong internet connectivity
[5]A.C. Doctor, “Integrated Educational Management Tool for Adamson University”	2022	Integrated university academic modules	Modular architecture, multi-module system	Automates attendance & academic records	Limited generalization across institutions
[6]GitHub, “Open- Source Department Portal Projects”	2018–2024	Open-source university portals	Analytics, personalization & automation	Shows trend toward intelligent personalization	Existing versions not fully scalable

Chapter 3

System Requirements and Specification

3.1 Hardware Requirements

- System Intel : Core i3 / i5 or above
- Processor : 2.0 GHz or higher, 64-bit recommended
- Hard Disk : Minimum 20 GB free space
- Monitor : 15” LED/LCD Display
- Input Devices : Standard Keyboard and Optical Mouse
- RAM : 8 GB recommended, 4 GB minimum
- Network : Stable Internet Connection for deployment

3.2 Software Requirements

- Operating System : Windows 10 / Windows 11 / Linux
- Programming Language : Python 3.10+
- Framework : Flask (Backend Web Framework)
- Frontend Technologies : HTML5, CSS3, JavaScript, Bootstrap 5
- Database : JSON-based storage
- IDE / Code Editor : Visual Studio Code (VS Code)
- Libraries : Flask, Jinja2, Werkzeug, Time, OS, JSON
- Tools : Git & GitHub (Version Control), Developer Tools
- Package Manager : pip (Python Package Installer)

Summary:

The proposed system, ISEHub – Student, Faculty, and Admin Academic Portal, is a lightweight Python-Flask web application requiring only moderate hardware and standard software tools. It follows a client–server architecture where users access the portal through a browser while the Flask backend manages all processing.

The system runs efficiently on an Intel i3/i5 machine with at least 4 GB RAM (8 GB preferred), around 20 GB of storage, a basic 15-inch monitor, and a stable internet connection for accessing APIs and uploaded files. Software requirements include Python 3.10+, Flask, HTML/CSS/Bootstrap, and JSON for storage, with development supported through VS Code, GitHub, and Postman. Flask’s minimal dependencies allow deployment on Windows, Linux, or macOS.

Overall, the specifications make ISEHub portable, scalable, and affordable for academic institutions, as it requires low hardware resources and uses open-source software without additional infrastructure costs.

Chapter 4

System Architecture and Implementation

4.1 Methodology

The methodology adopted for the development of the ISEHub – Academic Management Portal follows a structured, systematic, and iterative process to ensure reliability, usability, and smooth functioning across Admin, Student, and Faculty modules. The development approach is based on the Waterfall and Iterative Hybrid Model, ensuring both clear planning and flexible improvements during development.

The project began with requirement analysis, where the academic workflow of students and faculty was studied. Inputs from real-time academic activities—such as attendance management, internal marks entry, and study materials sharing—formed the foundation for understanding system needs. Based on these findings, the functional and non-functional requirements were documented.

In the system design phase, both high-level and low-level design diagrams were prepared. A client–server architecture was chosen, where the front-end acts as the client (HTML/CSS/Bootstrap) and Flask acts as the backend server. Data storage was handled using JSON files for lightweight operations, representing an easy-to-deploy mini-project solution. UI wireframes and flow diagrams were drawn to ensure user-friendly navigation for each module.

The implementation phase involved coding the Admin, Student, and Faculty modules in Flask. Each module was built as an independent but connected component with its own routes, templates, and data-handling functions. Features like user login, file uploads, attendance editing, and marks management were implemented using Python functions combined with HTML templates. Styling and animations were added to enhance user experience.

The system then proceeded to testing, where each page, form, and button was checked manually. Functional testing validated correct input handling, navigation, and data storage. Error handling was tested using invalid logins, empty submissions, and invalid uploads. Bugs were fixed iteratively to ensure smooth operation in real scenarios. Finally, integration testing verified that all modules (Admin, Student, Faculty) interact correctly.

The deployment phase completes the methodology, where the system is hosted locally and can later be deployed on a cloud platform. This ensures accessibility while keeping the setup simple for college demonstration and evaluation.

Summary:

The methodology adopted for developing the ISEHub Academic Management Portal followed a structured and systematic process to ensure efficiency and reliability in every stage of development. The work began with requirement analysis, where the challenges faced by students, faculty, and administrators in managing attendance, internal marks, study materials, and assignments were carefully studied. Based on these requirements, a clear system design was prepared, outlining the user flow, module interactions, and data handling structure. A JSON-based storage method was selected to support a lightweight, easily modifiable backend suitable for a mini-project. The frontend interface was then designed using HTML, CSS, and Bootstrap, focusing on a pastel-themed, user-friendly UI with role-specific dashboards for Admin, Faculty, and Students. Back-end development was carried out using Python Flask, implementing all routes, authentication mechanisms, file upload handlers, session control, and CRUD operations. Each module—attendance, internal marks, notes upload, notifications, assignment upload and student submissions—was coded with proper validation and data consistency checks. Integration of these modules ensured smooth communication between frontend actions and backend data updates. The portal was then tested thoroughly to confirm correct functioning of login flows, dashboard navigation, editing operations, and file handling processes. All bugs and inconsistencies identified during testing were fixed to improve robustness. The methodology concluded with UI refinements, optimization, and preparation for final deployment and demonstration, resulting in a fully functional, interactive academic management system.

4.1 System Design

The ISEHub System Architecture is structured in three interconnected layers to ensure smooth academic management. The User Interface Layer provides dedicated dashboards for Students, Faculty, and Admin, allowing each user to access only the features relevant to their role such as viewing materials, uploading notes, managing marks, or submitting assignments. The API & Application Layer, powered by the Flask backend, handles authentication, session management, and all core operations including file handling and JSON data updates. Finally, the Data Storage Layer consists of a JSON-based database for storing academic records and a server file storage system for handling uploaded notes, assignments, and submissions. Together, these layers enable secure, efficient, and user-friendly interaction across the entire portal. The following figure 4.1 shows the complete design of the proposed system that enables the students, faculties to upload, use and manage study materials along with the other functions like attendance etc.

ISEHUB SYSTEM ARCHITECTURE

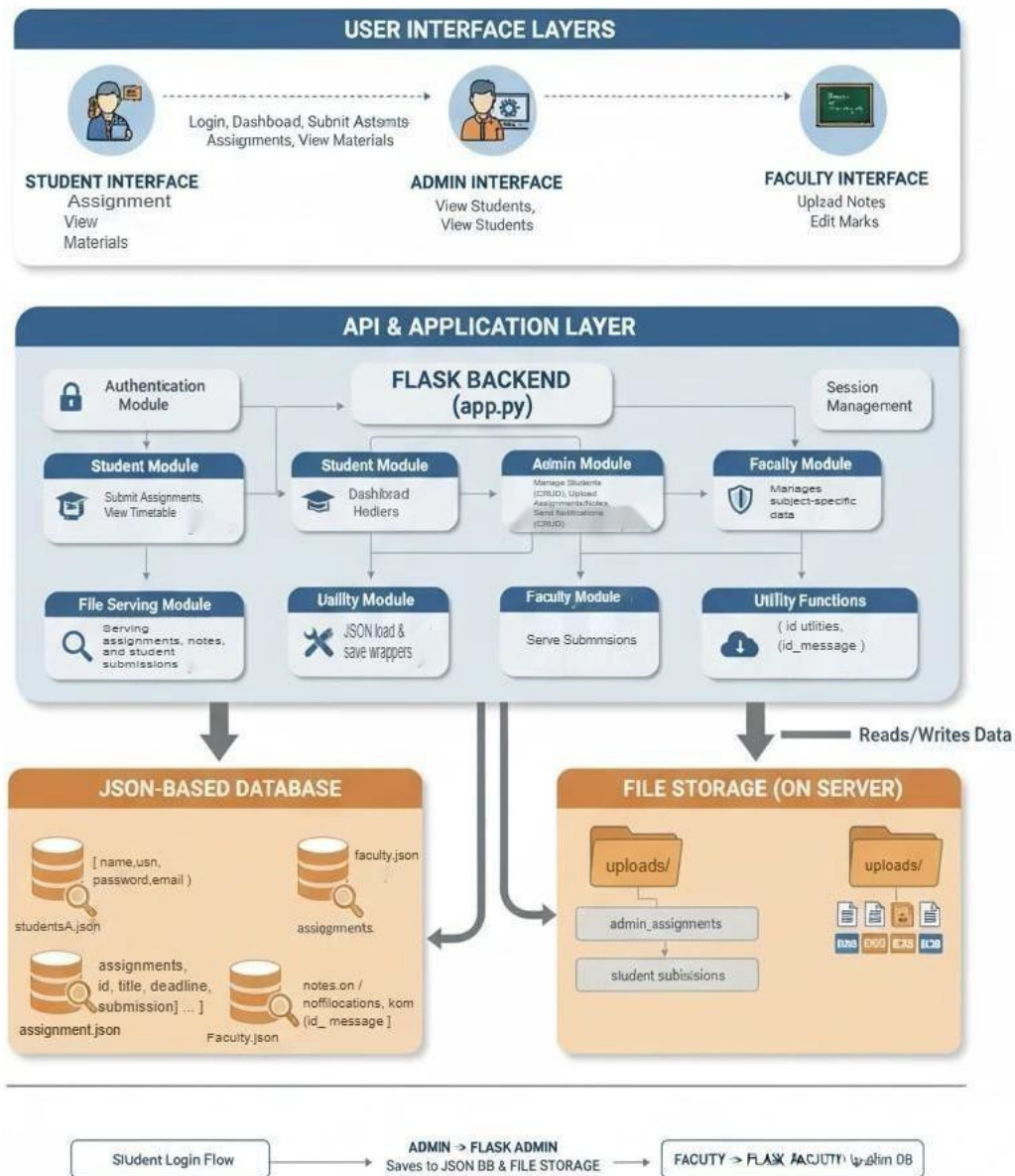


Figure 4.1 System Design

4.2 Data flow Diagram

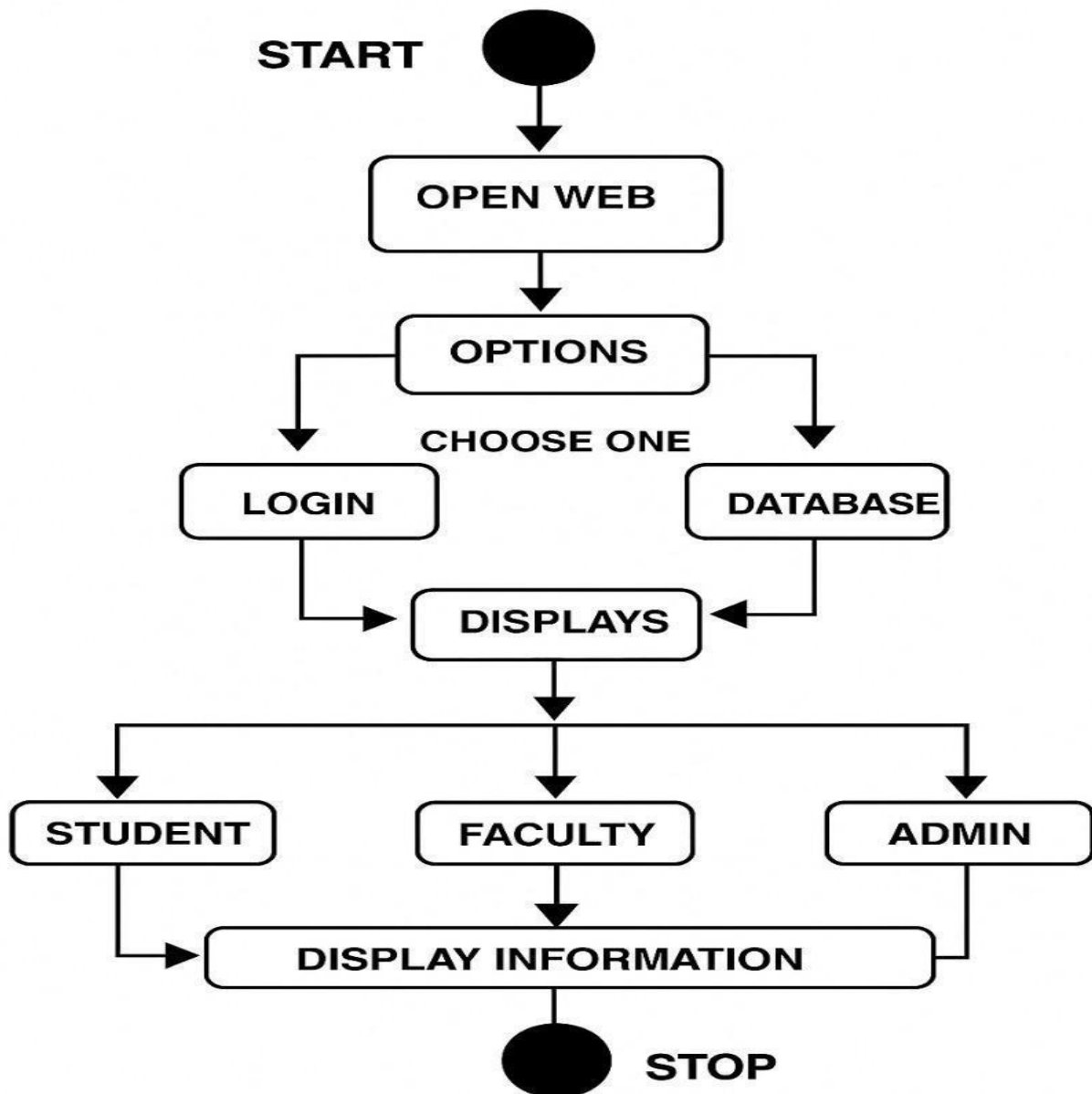


Figure 4.2 Data flow Diagram

This flowchart outlines the process for accessing a system. It starts by opening the web interface and presenting **OPTIONS** (Login or Database). Both paths lead to the **DISPLAYS** step, which bifurcates access based on user role: Student, Faculty, or Admin. The system then proceeds to **DISPLAY INFORMATION** relevant to the role before the process stops.

4.4 Pseudo Code

System Initialization

START

Load config (paths, file types)

Init session manager

Verify folders (students, notes, assignments, submissions)

Load datasets: students, faculty, assignments, notes, notifications

END

Helper Functions

allowed_file(filename) → TRUE if extension allowed

save_file(file, folder) → save with timestamped safe name → return filename

load_students(section) / save_students(section, data)

find_student(identifier) → search across sections → return student, section

load_json(file) / save_json(file, data)

Authentication

POST /login:

If admin credentials → open admin dashboard

Else if student credentials → open student dashboard

Else if faculty credentials → open faculty dashboard

Else → invalid credentials

GET /logout:

Clear session → redirect login

Admin Module

/admin/dashboard → tiles for students, assignments, notes, notifications, attendance, marks, submissions

/admin/students/<section> → list students

/admin/internal_marks/<usn> → view/update 6 subjects

/admin/attendance/<usn> → update attendance

/admin/notes (GET/POST) → upload/delete study materials

/admin/notifications (GET/POST) → add/delete notifications

Student Module

/student/dashboard → tiles: assignments, notes, attendance, marks, notifications, profile

/study_material → view and download notes

/student/assignments/<id> → view & upload submission

/internal_marks → view marks

/attendance → view attendance chart

/notifications → view announcements

/profile → view personal details

Faculty Module

/faculty/dashboard → subjects + sections handled

/faculty/students → list students assigned

/faculty/attendance/<usn>/<subject> → update attendance

/faculty/marks/<usn>/<subject> → update marks

/faculty/notes (GET/POST) → upload/view notes

File Download

/files/<category>/<filename> → validate folder → download file

Error Handling

404 → page not found

Unauthorized → redirect login

Invalid file upload → show error

System Shutdown

On server stop → safely save datasets

END

Chapter 5

Results and Discussions

5.1 Login Screen

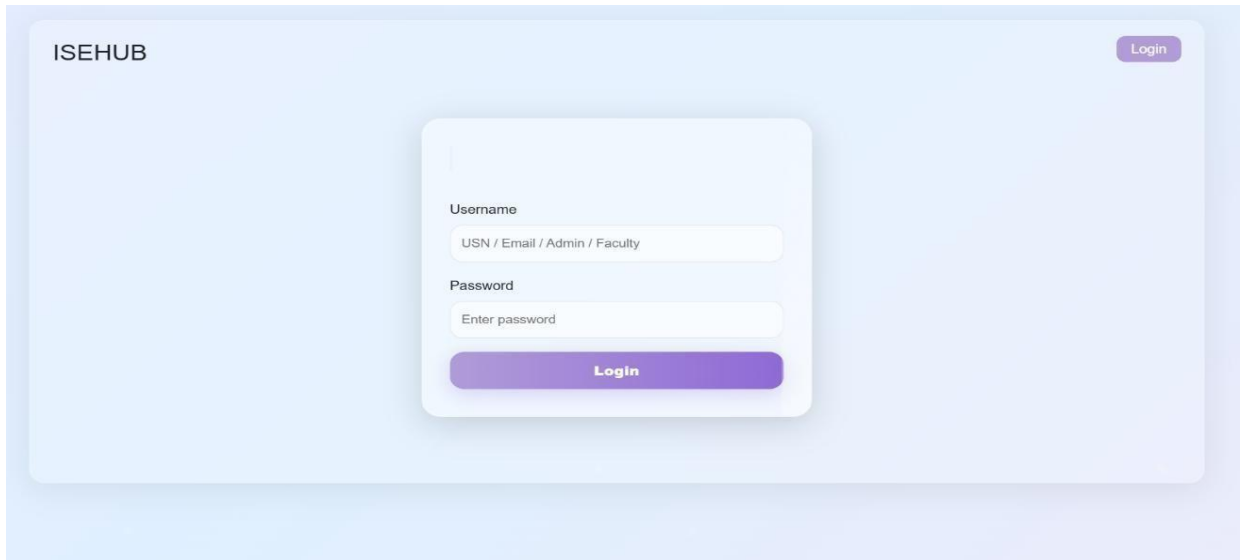


Figure 5.1 Login Screen

Login screen displaying a clean pastel-themed interface with username and password fields.

5.2 Logout Screen

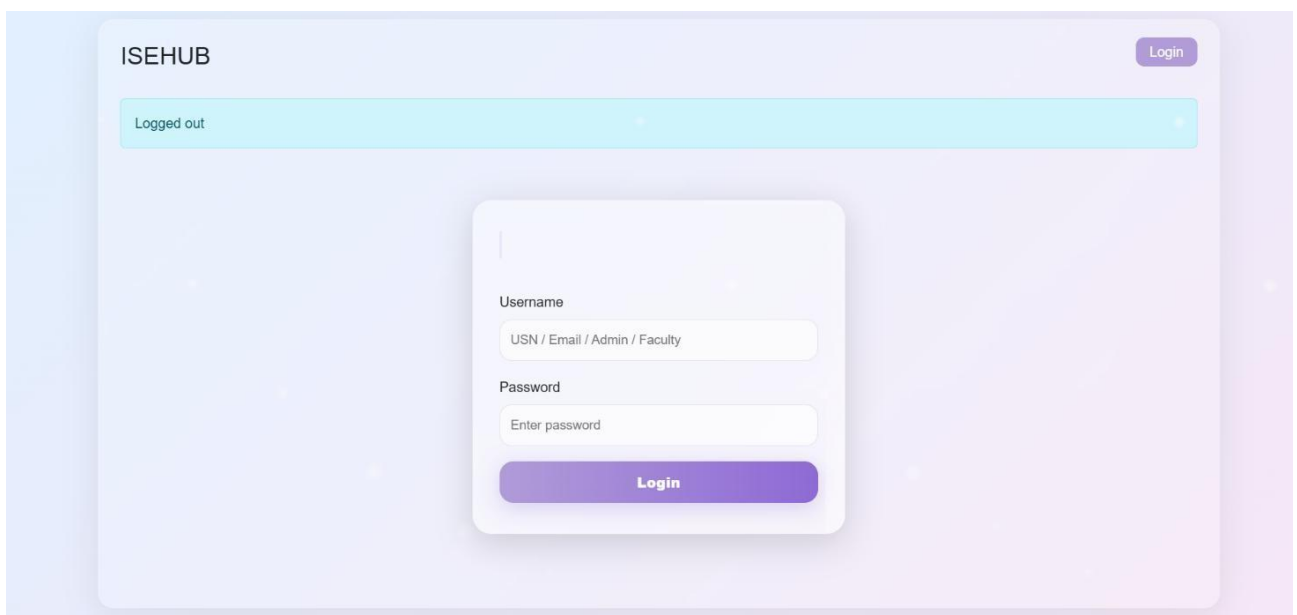


Figure 5.2 Logout Screen

Logout screen displaying a “Logged out” message above the login form.

5.3 Admin Dashboard

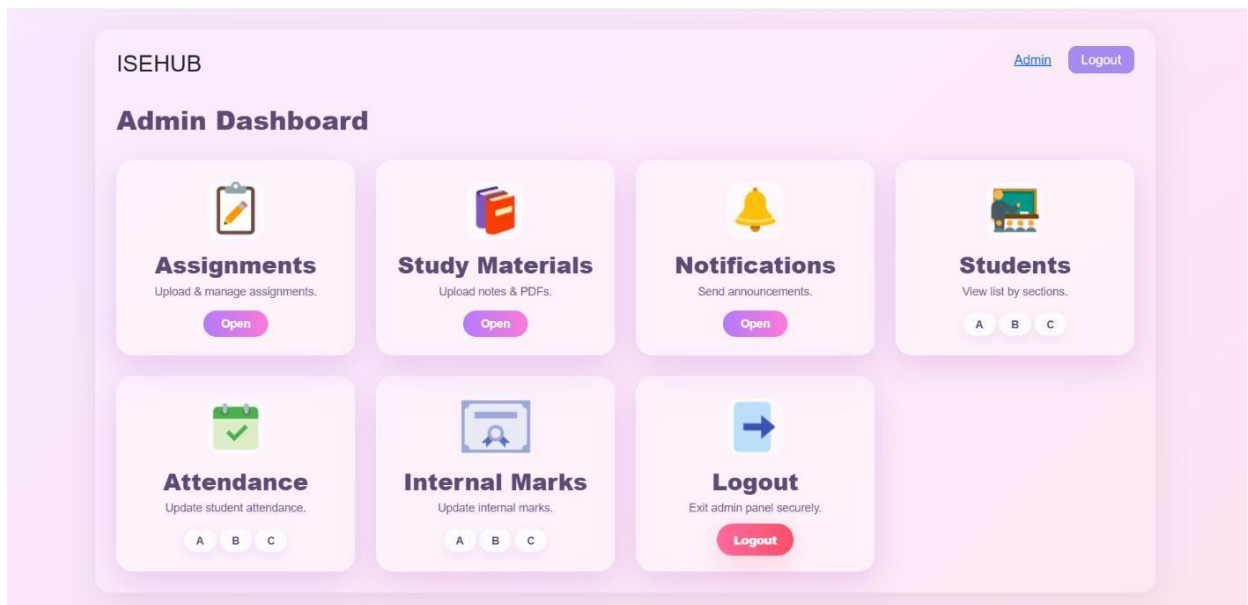


Figure 5.3 Admin Dashboard

Admin dashboard displaying quick-access cards for assignments, study materials, notifications, students, attendance, internal marks, and logout.

5.4 Faculty Dashboard

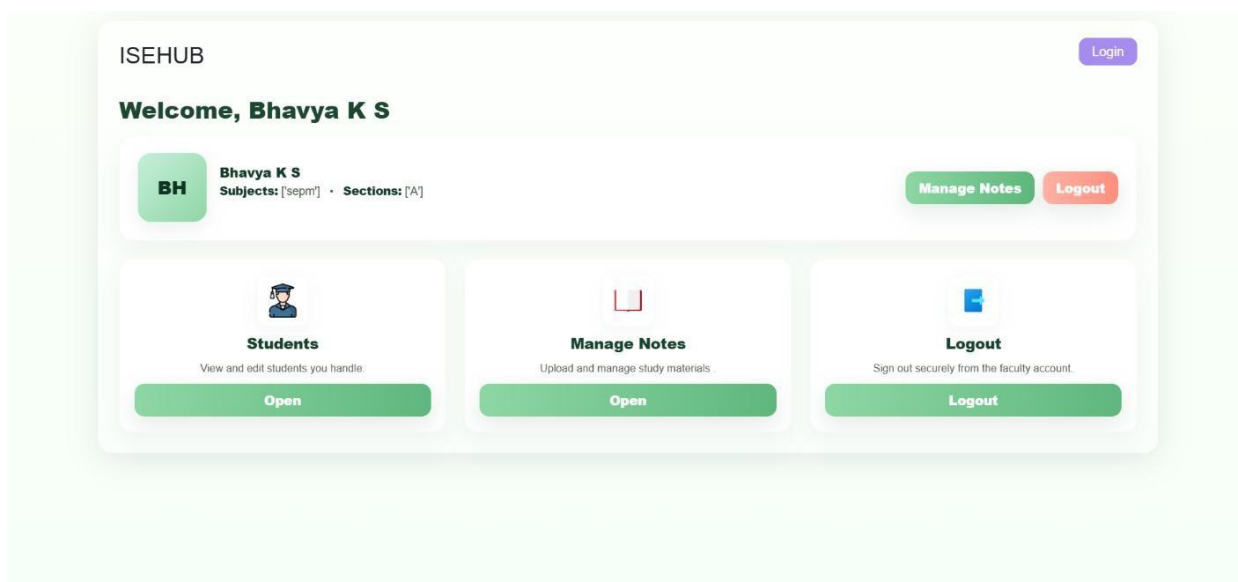


Figure 5.4 Faculty Dashboard

Faculty dashboard showing user details and quick-access options for managing students, notes, and logout.

5.5 Student Dashboard

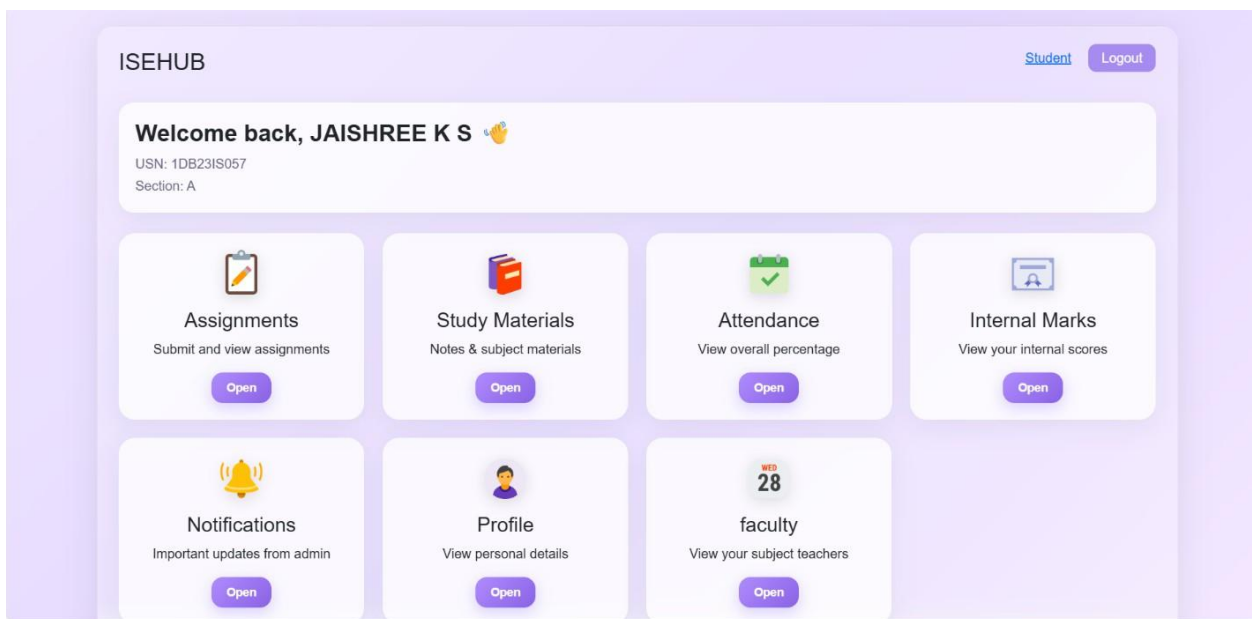


Figure 5.5 Student Dashboard

Student dashboard displaying quick-access tiles for assignments, study materials, attendance, internal marks, notifications, profile, and faculty details.

Chapter 6

Advantages , Disadvantages and Applications

6.1 Advantages

- Centralized storage of all academic data.
- Reduces manual work for admin and faculty.
- Minimizes human errors in attendance and marks.
- Easy to use because of clean dashboard UI.
- Secure role-based login system (Admin/Faculty/Student)
- Promotes paperless documentation.
- Accessible anytime from any device with internet.
- Improves transparency between faculty and students

6.2 Disadvantages

- Requires constant internet connection for use.
- No offline data entry or access.
- Users may need basic training to use the system.
- Dependent on password security.
- JSON-based storage may slow down with large data.
- Needs regular backup to avoid file loss.
- Limited error handling if files are corrupted.
- Cannot function properly if admin mismanages data.

6.3 Applications

▪ Centralized Academic Resources:

The system provides a unified platform where students can easily access notes, assignments, question papers, and study materials uploaded by faculty. This ensures quick availability of academic content without dependency on manual distribution.

▪ Internal Marks & Attendance Tracking:

Students can instantly check their internal marks and attendance percentage through the portal. This helps them monitor their academic progress regularly and take corrective steps when required.

▪ Faculty Academic Management:

Faculty members can upload study materials, manage assignments, update marks, and view their allocated sections through a structured and secure interface. This reduces paperwork and simplifies routine academic tasks.

▪ Department Notifications & Announcements:

The portal broadcasts important department-level updates such as exam schedules, events, holidays, and circulars. This eliminates delays caused by manual communication and ensures everyone receives timely information.

▪ Student Information Management:

The system maintains structured data of student profiles, including personal information, USN, academic records, and course details. This allows the department to access accurate information quickly whenever needed.

Summary:

The Smart Department Portal simplifies academic management by bringing together key functions such as attendance, internal marks, study materials, notifications, and student records into one digital platform. Its applications ensure faster communication, easier access to information, and reduced manual work for students, faculty, and administrators. The portal improves transparency, saves time, and enhances overall departmental efficiency.

Conclusion and Future Scope

Conclusion:

The Smart Department Portal effectively digitalizes key academic processes, making attendance, marks, notifications, and study material management faster and more accurate. The system enhances communication and accessibility by providing students and faculty with a centralized platform to interact and access essential information anytime. Overall, the portal improves departmental efficiency and transparency, offering a scalable solution that supports modern academic needs and future enhancements.

Future Scope:

- **AI-Based Performance Prediction:** Implement machine learning models to predict student performance, attendance risks and academic outcomes.
- **Mobile Application Integration:** Develop Android and iOS apps for easier access to attendance, notes, marks, and notifications.
- **Biometric / RFID Attendance System:** Integrate fingerprint, face recognition, or RFID-based smart ID cards to automate attendance tracking.
- **Cloud Deployment:** Deploy the system on cloud platforms (AWS, Azure, GCP) for scalability, reliability, and multi-campus access.
- **Chatbot Assistance:** Add an AI chatbot to help students with FAQs, notes, exam schedules, and academic queries.
- **Enhanced Security Features:** Implement advanced authentication such as OTP login, 2FA, and encrypted data storage.
- **Placement & Internship Module:** Extend the system to include resume uploads, placement notifications, company profiles, and student tracking.

References

- [1] “Web Portals for Higher Education Institutions: A Review,” International Journal of Information Management, 2018. This paper reviews web portals used in higher education and explains how they improve information access, student interaction, and administrative efficiency.
- [2] S. Pavithra et al., “Department Information System – Web Application,” IJERT, vol. 8, no. 5, 2019. The authors developed a departmental information system that automates key activities such as managing student records, faculty information, and department notices.
- [3] I. Z. Kogotkova et al., “Smart Department: Designing Digital Applications for Smart Organizations,” Springer, 2021. This research focuses on designing smart departmental applications that support automation, real-time data handling, and smart campus integration.
- [4] F. M. Gonzalez-Longatt et al., “Secured Student Portal Using Cloud,” Expert Clouds and Applications, 2021. The study proposes a secure, cloud-based student portal that ensures data protection while offering scalable access to academic services.
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- [6] GitHub, “Open-Source Department Portal Projects,” Available: <https://github.com>. This source includes open-source departmental and student portal systems useful as reference codebases for academic projects.